25901006 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCZIM102 - FUNDAMENTALS OF BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer **All** Questions

Each Question carries **1** mark

1. Cosmozic theory is also known as…… [U] / [CO1]
2. Passive diffusion means it does not require ..... in order to occur. [U] / [CO3, CO4]
3. During symbiotic nitrogen fixation cellwall of roothair invaginates to form..... [K] / [CO5]
4. .....,an italian physicist first identified golgi complex [K] / [CO4]
5. CAM Plants carry out carbondioxide fixation at ..... [U] / [CO5]
6. In the field of pharmaceuticals ,colloids are used to form .....that help deliver drugs in a colloidal manner. [U] / [CO2]
7. The pOH scale measures the concentration of ..... in a solution. [U] / [CO2]
8. Membrane lipids are .....in nature. [U] / [CO3]

9. Prokaryotes have .....ribosomes ,each consist of 30 S and 50 S subunits. [K] / [CO4]

10. Phenolics are a class of secondary metabolites in plants that are characterized by the presence of one or more.....ring [K] / [CO5]

[10x1 = 10]

Part B

Short Questions

Answer any **10** Questions

Each Question carries **2** marks

11. Explain the role of grana in photosynthesis. [U] / [CO5]

12. Define Covalant bond. [U] / [CO2]

13. Define ATP [A] / [CO4]

14. Define autotrophs. [K] / [CO5]

15. List out the main events occur during the stage of mitosis. [K] / [CO4]

16. Write the Equation for the ionic product of water. [U] / [CO2]

17. List out the electron acceptors present in cyclic and non cyclic photophosphorylation. [U] / [CO5]

18. State Lock & Key Model of enzyme. [K] / [CO1]

19. Describe the importance of buffer capacity in biological system. [U] / [CO2]

20. Write a note on protein present in plasma membrane. [A] / [CO3]

21. Define Isotonic solution. [K] / [CO4]

22. Summarize the differences between atmospheric nitrogen and fixed nitrogen. [U] / [CO5]

[10x2 = 20]

Part C

Short Essay Type Questions

Answer any **5** Questions

Each Question carries **4** marks

23. Describe the the molecular mechanism involved in Non symbiotic nitrogen fixation. [A] / [CO5]

24. Analyze tthe impact of lysosomal dysfunction on cellular process [An] / [CO4]

25. Differentiate lyophyllic colloid from lyophobic colloids. [U] / [CO2]

26. Describe the structure and function of Endoplasmic Reticulum. [An] / [CO4]

27. Outline the importance of hemoglobin buffer system in maintaining blood pH. [U] / [CO2]

28. Distinguish between endocytosis and exocytosis. [K] / [CO4]

29. Compare and contrast C3 pathway and C4 pathway. [A] / [CO5]

[5x4 = 20]